

A Positive- γ Counterexample to the Nonlocal Poincaré Question as Stated

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Status. Candidate counterexample, likely valid. It gives a complete negative answer to the positive- γ branch under the natural interpretation that the displayed estimate is uniform for all $\lambda > 0$. The source does not print that quantifier explicitly; a sufficiently-large- λ repair is not addressed.

1 Source question and normalization

The source is Hoai-Minh Nguyen, *Characterizations of the Sobolev norms and the total variation via nonlocal functionals, and related problems*, arXiv:2507.00842v2 [1]. Open Question 3 appears at the end of Section 9 on PDF page 25.

In this direction, concerning Φ_λ , we ask the following question viewing Proposition [2.1](#)

Open question 3. Let Q be a unit cube of $\mathbb{R}^N$, and let $p \geq 1$, $\gamma \in \mathbb{R} \setminus (-1, 0]$ be such that $\kappa_{N,p,\gamma} > 0$. Is it true that

$$\iint_{Q \times Q} |u(x) - u(y)|^p dx dy \leq C_{N,p} \left(\Phi_\lambda(u) + \lambda^p \right) \text{ for } u \in L^p(Q)?$$

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For a unit cube $Q \subset \mathbb{R}^N$, the domain-correct normalized local functional is

$$\Phi_{\lambda,Q}(u) := \lambda^p \iint_{\substack{Q \times Q \\ |u(x) - u(y)| > \lambda |x - y|^{1+\gamma/p}}} |x - y|^{-N+\gamma} dx dy. \quad (1)$$

The question asks, for $p \geq 1$ and $\gamma \in \mathbb{R} \setminus (-1, 0]$ with $\kappa_{N,p,\gamma} > 0$, whether

$$\iint_{Q \times Q} |u(x) - u(y)|^p dx dy \leq C_{N,p} (\Phi_{\lambda,Q}(u) + \lambda^p), \quad u \in L^p(Q). \quad (2)$$

The source displays a free λ and a constant independent of λ ; we use the standard uniform-in- $\lambda > 0$ reading. Nguyen writes $\Phi_\lambda(u)$ in the question even though u is defined only on Q . Equation (1) is the local functional defined in the paper. A global definition in the TeX source accidentally omits the factor λ^p , but the local definition, Theorem 1.3, and all later calculations use this factor.

2 Proof intuition

For $\gamma > 0$ the kernel $|x - y|^{-N+\gamma}$ is integrable across the diagonal on a bounded cube. Consequently the entire normalized nonlocal energy is at most a finite constant times λ^p , no matter what u is. Sending λ to zero therefore erases the proposed right-hand side, while the mean oscillation of any nonconstant u remains positive.

There is one potential logical escape: perhaps no positive γ satisfies the hypothesis $\kappa_{N,p,\gamma} > 0$. Gobbino and Picenni's Γ -liminf theorem [2] closes that escape. A clipping argument converts Nguyen's convergence-in-measure cell problem to their L^1 topology without increasing the energy, proving that the cell constant is positive for every $\gamma > 0$.

3 The counterexample

Theorem 1 (Positive- γ branch fails). *Let $N \geq 1$, $p \geq 1$, $\gamma > 0$, and $Q = (0, 1)^N$. Then $\kappa_{N,p,\gamma} > 0$, but there is no finite constant C for which (2) holds for every $u \in L^p(Q)$ and every $\lambda > 0$. Indeed the assertion fails for $u(x) = x_1$. This remains true if C is allowed to depend on N, p, γ , and Q .*

We separate verification of the premise from the short counterexample.

Proposition 1 (The positive- γ premise is nonvacuous). *For every $N \geq 1$, $p \geq 1$, and $\gamma > 0$, Nguyen's cell constant satisfies*

$$\kappa_{N,p,\gamma} \geq A_{N,p} \frac{\log 2}{2^{\gamma+1} - 1} > 0, \quad (3)$$

where $A_{N,p} = \int_{\mathbb{S}^{N-1}} |v \cdot \omega|^p d\mathcal{H}^{N-1}(\omega) > 0$ for any unit vector v .

Proof. In Nguyen's definition, one takes the infimum of $\liminf_{\lambda \rightarrow \infty} \Phi_{\lambda,Q}(h_\lambda)$ over measurable families converging in measure on Q to

$$h(x) = \frac{x_1 + \cdots + x_N}{\sqrt{N}}.$$

Fix such a family and let $T(t) = \min\{\max\{t, 0\}, \sqrt{N}\}$. Since T is 1-Lipschitz and $T(h) = h$, its exceedance set is contained in that of the original function:

$$\Phi_{\lambda,Q}(T \circ h_\lambda) \leq \Phi_{\lambda,Q}(h_\lambda).$$

Moreover $T \circ h_\lambda \rightarrow h$ in measure. The functions are uniformly bounded on the finite-measure cube, so this convergence is also in $L^1(Q)$.

Gobbino–Picenni's Theorem 1.1 [2], applied on the open cube, now gives

$$\liminf_{\lambda \rightarrow \infty} \Phi_{\lambda,Q}(T \circ h_\lambda) \geq A_{N,p} \frac{\log 2}{2^{\gamma+1} - 1} F_p(h, Q).$$

Here $F_p(h, Q) = 1$: the gradient of h has Euclidean norm one and $|Q| = 1$ (the same computation is the total variation when $p = 1$). Combining the two inequalities and taking the infimum over all families proves (3). $\square$

Proof of Theorem 1. Set $u(x) = x_1$. By Fubini,

$$\begin{aligned} \iint_{Q \times Q} |u(x) - u(y)|^p dx dy &= \int_0^1 \int_0^1 |s - t|^p ds dt \\ &= 2 \int_0^1 \int_0^s (s - t)^p dt ds = \frac{2}{(p+1)(p+2)} > 0. \end{aligned} \quad (4)$$

On the other hand, because $Q - Q \subset B_{\sqrt{N}}(0)$ and $\gamma > 0$,

$$\begin{aligned} \nu_\gamma(Q \times Q) &= \iint_{Q \times Q} |x - y|^{-N+\gamma} dx dy \\ &\leq |\mathbb{S}^{N-1}| \int_0^{\sqrt{N}} r^{\gamma-1} dr = \frac{|\mathbb{S}^{N-1}|}{\gamma} N^{\gamma/2} < \infty. \end{aligned} \tag{5}$$

The exceedance set in (1) is contained in $Q \times Q$, so

$$0 \leq \Phi_{\lambda,Q}(u) \leq \frac{|\mathbb{S}^{N-1}|}{\gamma} N^{\gamma/2} \lambda^p. \tag{6}$$

Thus $C(\Phi_{\lambda,Q}(u) + \lambda^p) \rightarrow 0$ as $\lambda \downarrow 0$ for every finite C , contradicting the fixed positive quantity (4). The preceding proposition verifies the required hypothesis $\kappa_{N,p,\gamma} > 0$. $\square$

4 Global zero-extension robustness

Although the local functional is the natural reading, the obstruction also survives a normalized global interpretation. Extend $u(x) = x_1$ on Q by zero and call the result U on $\mathbb{R}^N$. If (x, y) belongs to its exceedance set, at least one endpoint lies in Q , $|U(x) - U(y)| \leq 1$, and therefore

$$|x - y| < R_\lambda := \lambda^{-p/(p+\gamma)}.$$

Double-counting which endpoint lies in Q gives

$$\begin{aligned} \Phi_\lambda(U) &\leq 2\lambda^p \int_Q \int_{|x-y| < R_\lambda} |x - y|^{-N+\gamma} dy dx \\ &= \frac{2|\mathbb{S}^{N-1}|}{\gamma} \lambda^p R_\lambda^\gamma = \frac{2|\mathbb{S}^{N-1}|}{\gamma} \lambda^{p^2/(p+\gamma)} \rightarrow 0. \end{aligned}$$

Hence zero extension does not rescue the estimate.

5 Verification, scope, and novelty

The argument is analytic and exact. It checks the premise separately, gives an explicit positive left-hand side, and bounds the entire energy rather than approximating its exceedance set. It also proves the stronger statement that no additive remainder $r(\lambda) \rightarrow 0$ can make a uniform inequality of this form true for positive γ .

The conclusion is deliberately scoped. It answers the positive- γ branch of Open Question 3 under the uniform reading forced by the free parameter in the display. It does not address $\gamma \leq -1$. If the intended question was instead restricted to all sufficiently large λ when $\gamma > 0$, the present small- λ argument does not apply.

A bounded novelty search on 11 August 2026 covered the run indexes, the exact arXiv id and title, exact-phrase searches for Open Question 3, searches for “nonlocal Poincaré”, Φ_λ , and $\kappa_{N,p,\gamma}$, and the source paper’s references. No later paper explicitly stating this counterexample or resolving Open Question 3 was found. This supports but does not certify novelty. Human review should focus on the intended λ quantifier and the source’s local/global notation.

References

- [1] H.-M. Nguyen, *Characterizations of the Sobolev norms and the total variation via nonlocal functionals, and related problems*, arXiv:2507.00842v2, 2025.
- [2] M. Gobbino and N. Picenni, *Gamma-liminf estimate for a class of non-local approximations of Sobolev and BV norms*, J. Funct. Anal. **289** (2025), no. 9, Paper 111106; arXiv:2311.05560.